Unraveling the Spectro-polarimetric Difference of Repeating and Non-repeating FRBs

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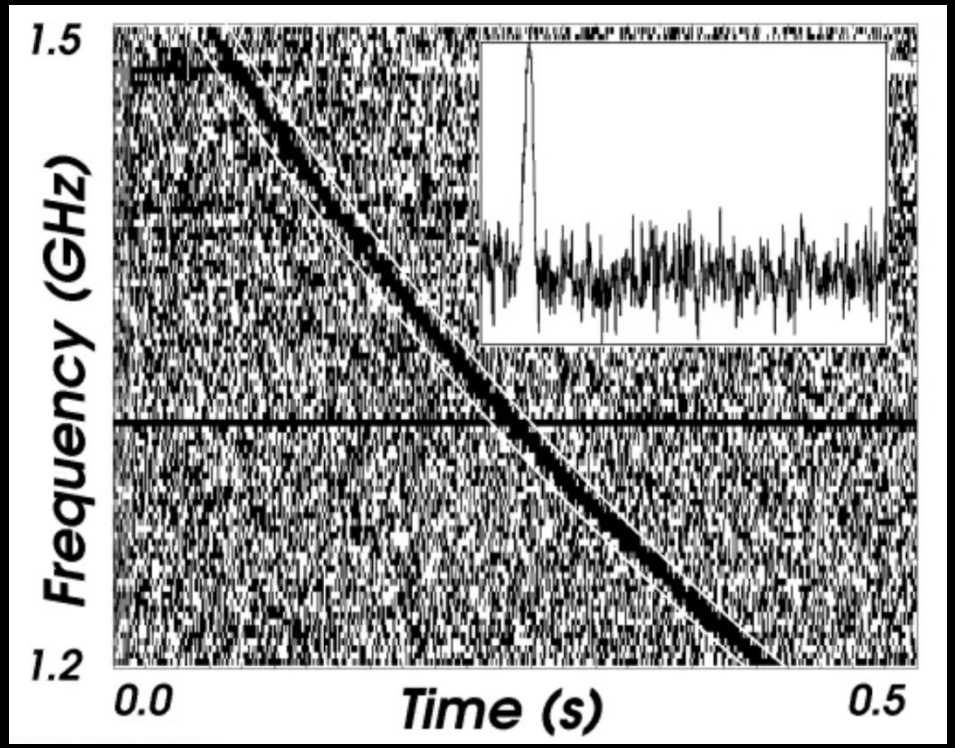
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Background

- Bright millisecond-duration radio transients
- First discovered in 2007 (Lorimer burst)
- First repeating source: FRB 20121102



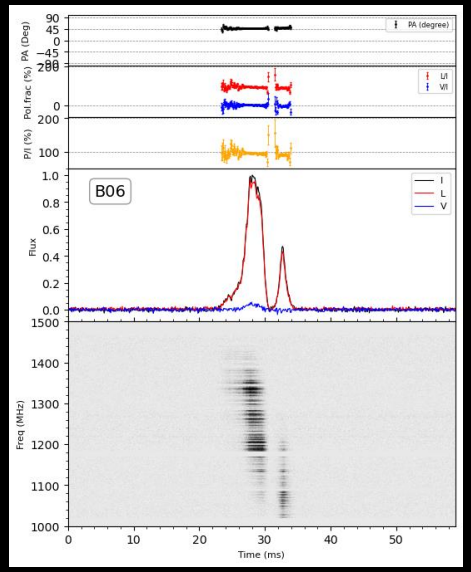
D. R. Lorimer et al. 2007

- **Are repeating and non-repeating FRBs intrinsically different populations?**

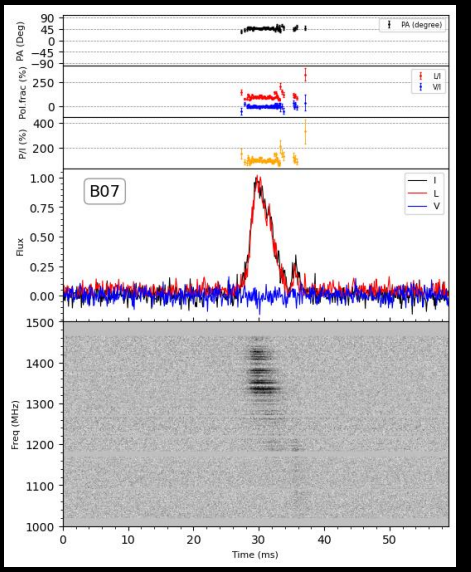
Spectra

- Repeating FRBs show rich temporal–spectral structures
  - Downward frequency drifting sub-bursts
  - Narrow-band emission features
  - Suggest structured emission process

→ **Indicate possible physical differences from one-off bursts**



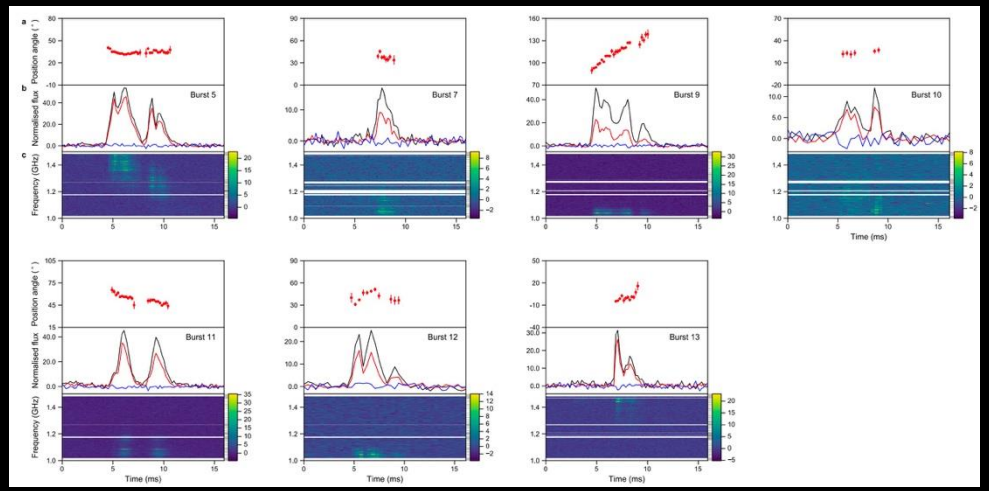
Drift



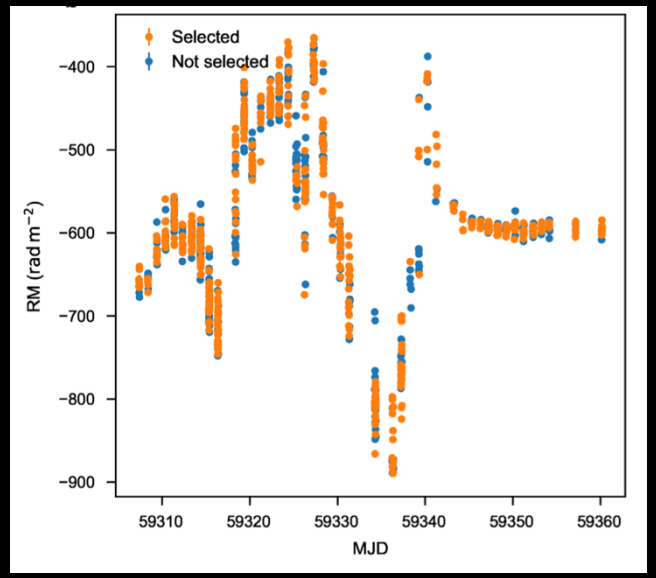
Narrow-band emission

Polarization

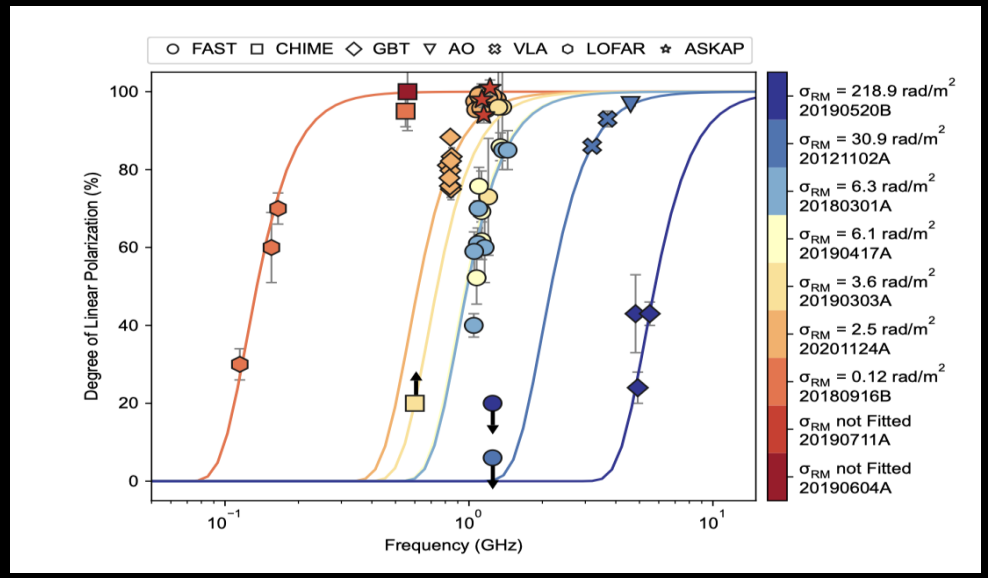
- Polarization probes emission mechanisms and magnetic environments



Luo et al. 2020



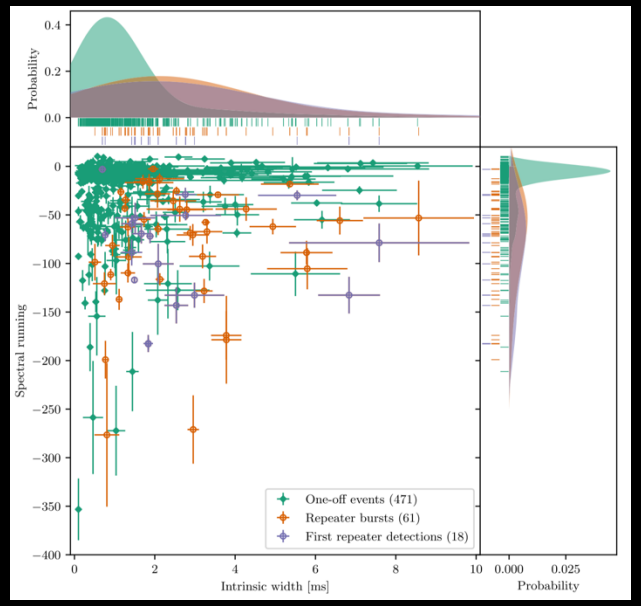
Xu et al. 2022



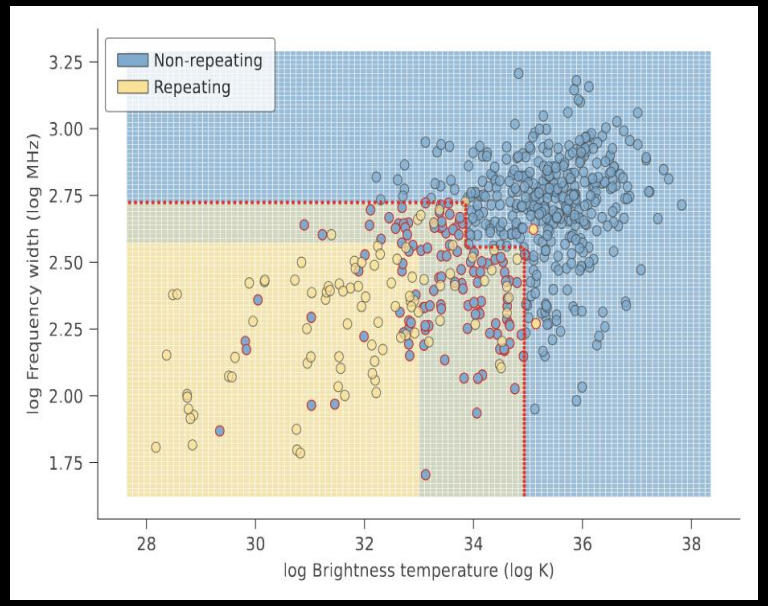
Feng et al. 2022

Previous Work

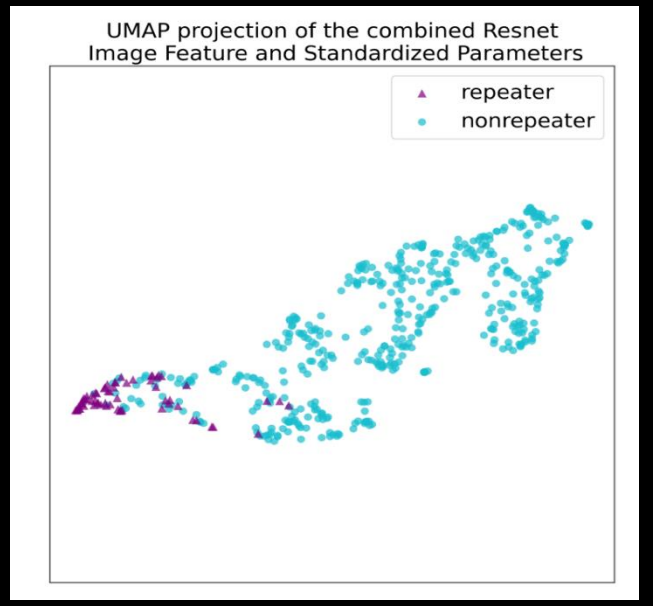
- Previous studies have evolved from statistical analysis to deep learning.



Pleunis et al. 2021



Luo et al. 2022



yang et al. 2023

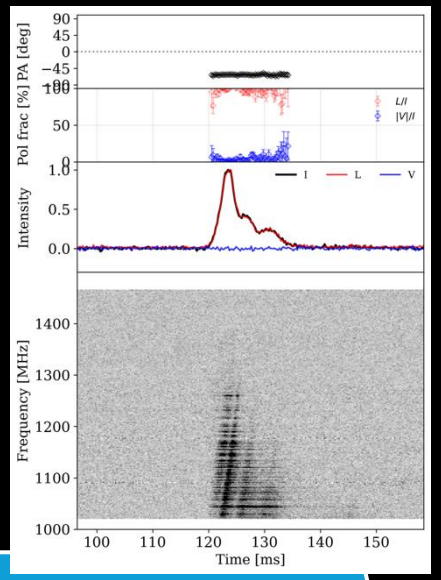
Motivation

- Spectra: Time-frequency structure

- PA evolution: Emission geometry and magnetized environment

- **Method:**
  1D Convolutional Neural Network
  Mechine Learning: Dimensionality Reduction, Clustering and Classification.

Data Set

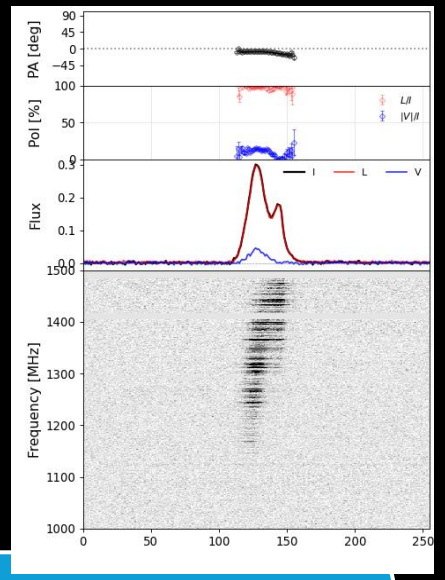


Repeating FRBs:

536 bursts from

FRB 20201124A observed by FAST

Jiang et al. 2024

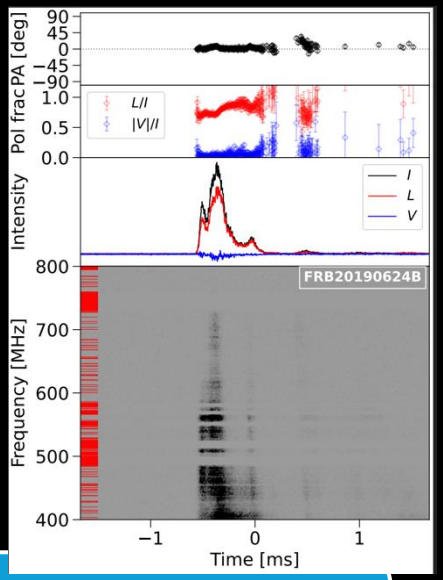


Repeating FRBs:

700 bursts from

FRB 20220912A observed by FAST

Zhang et al. 2023



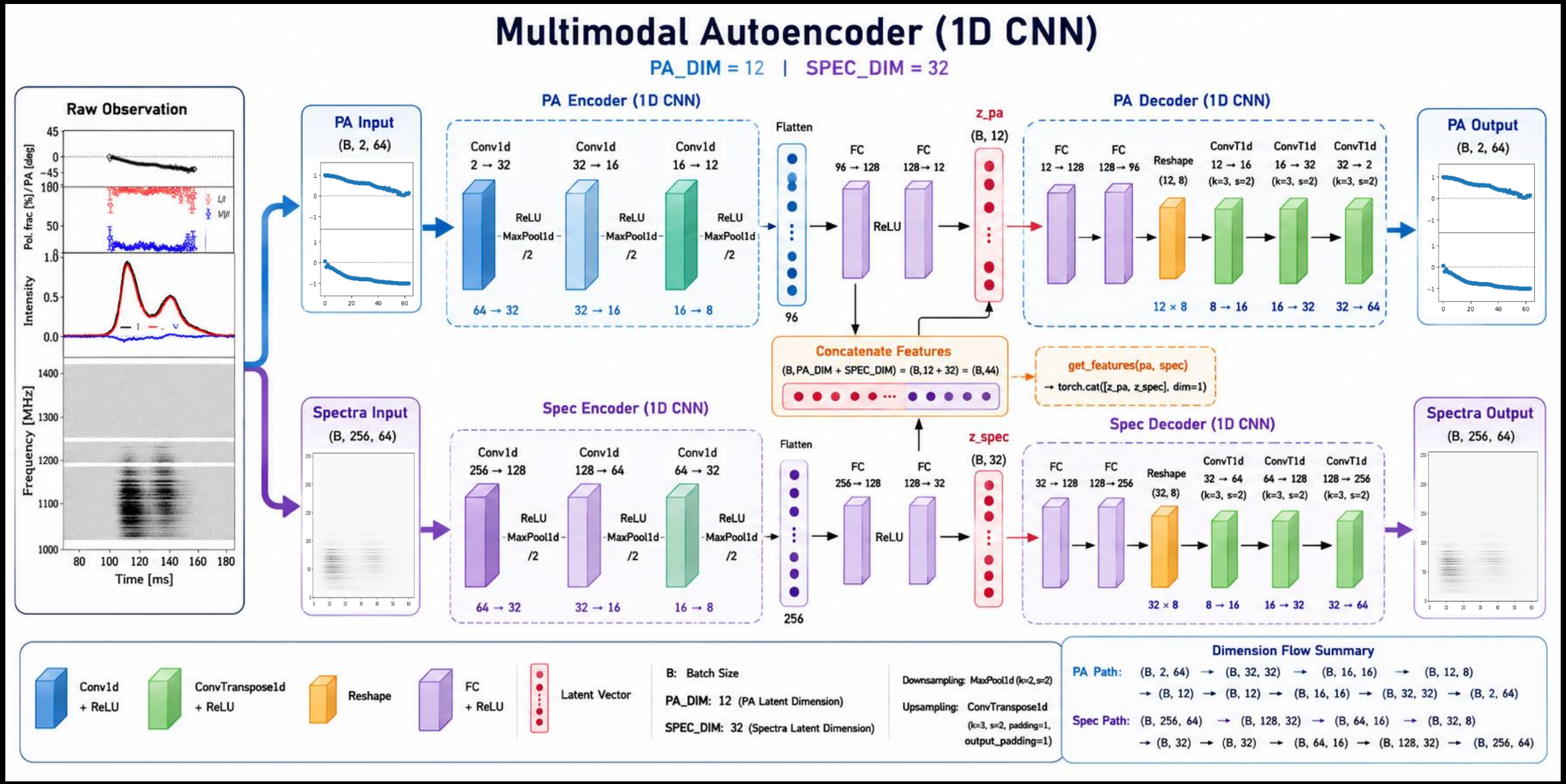
Non-repeating FRBs:

64 bursts from

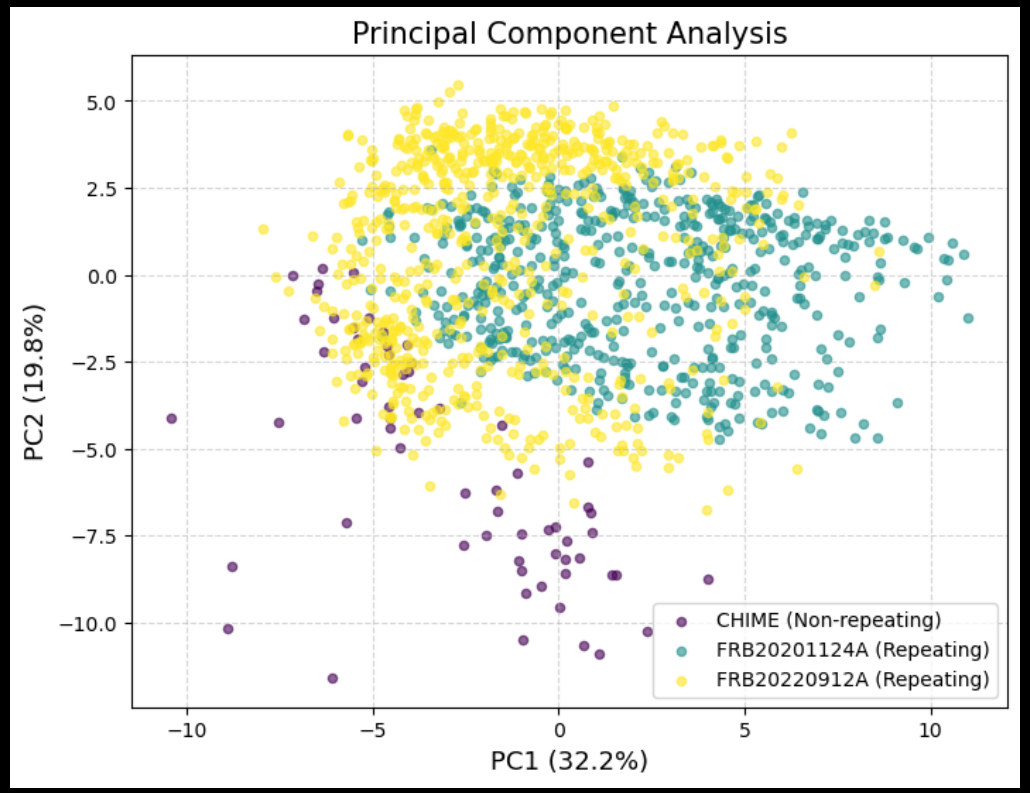
the first CHIME/FRB Baseband catalog

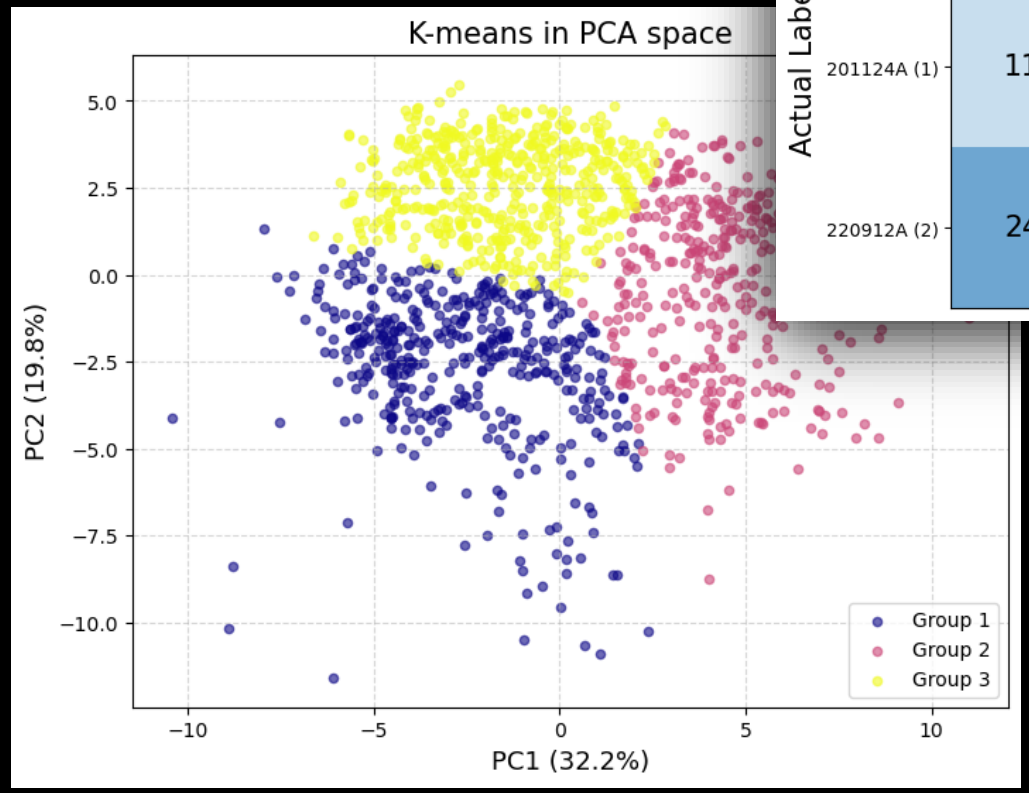
Pandhi et al. 2024

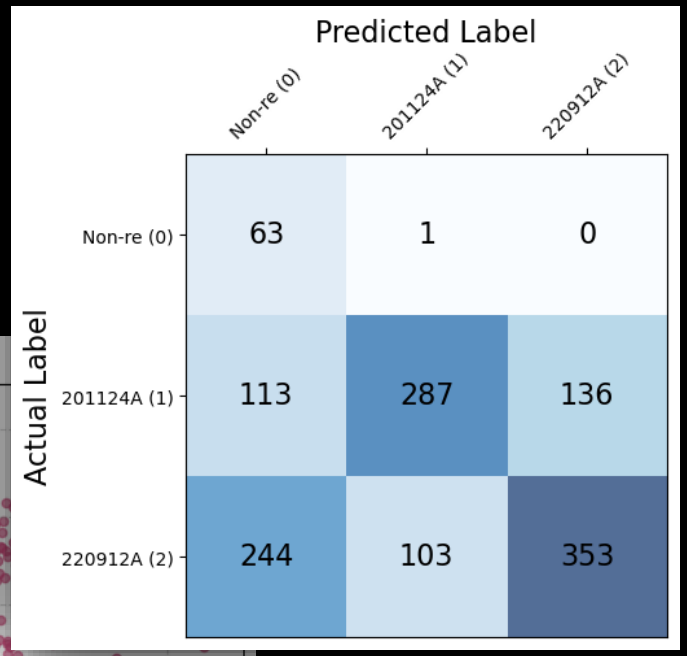
Features (1D CNN)



PCA+K-Means

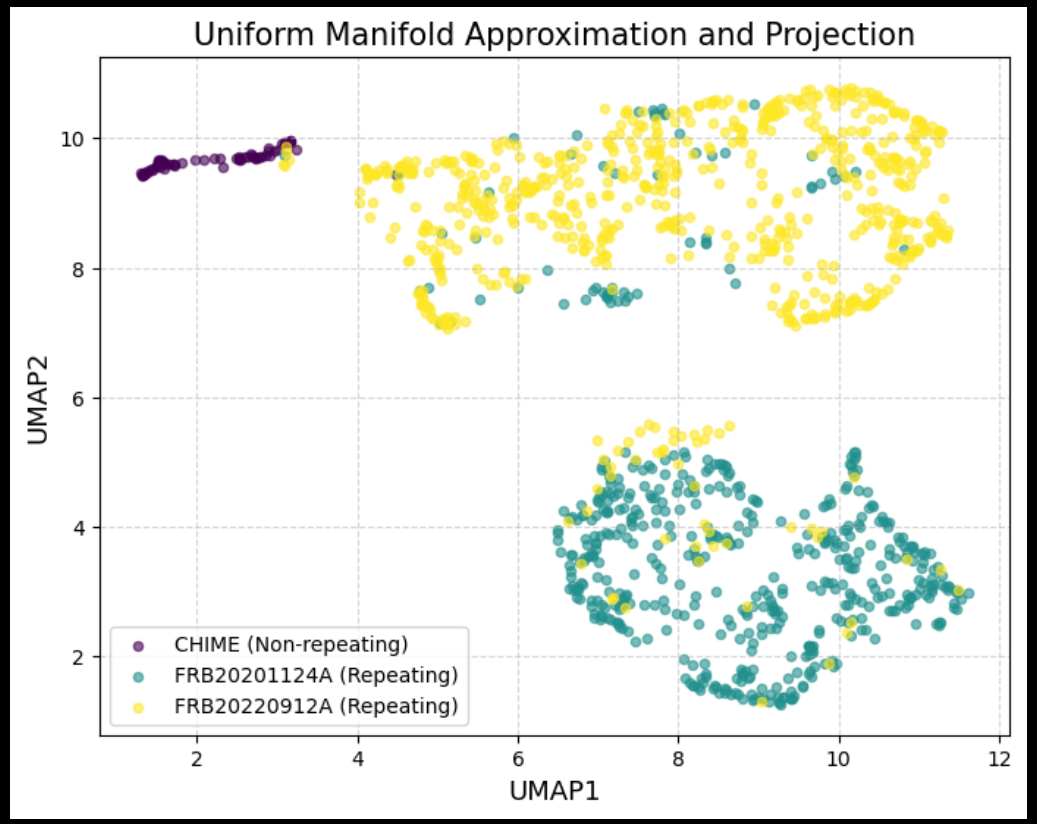


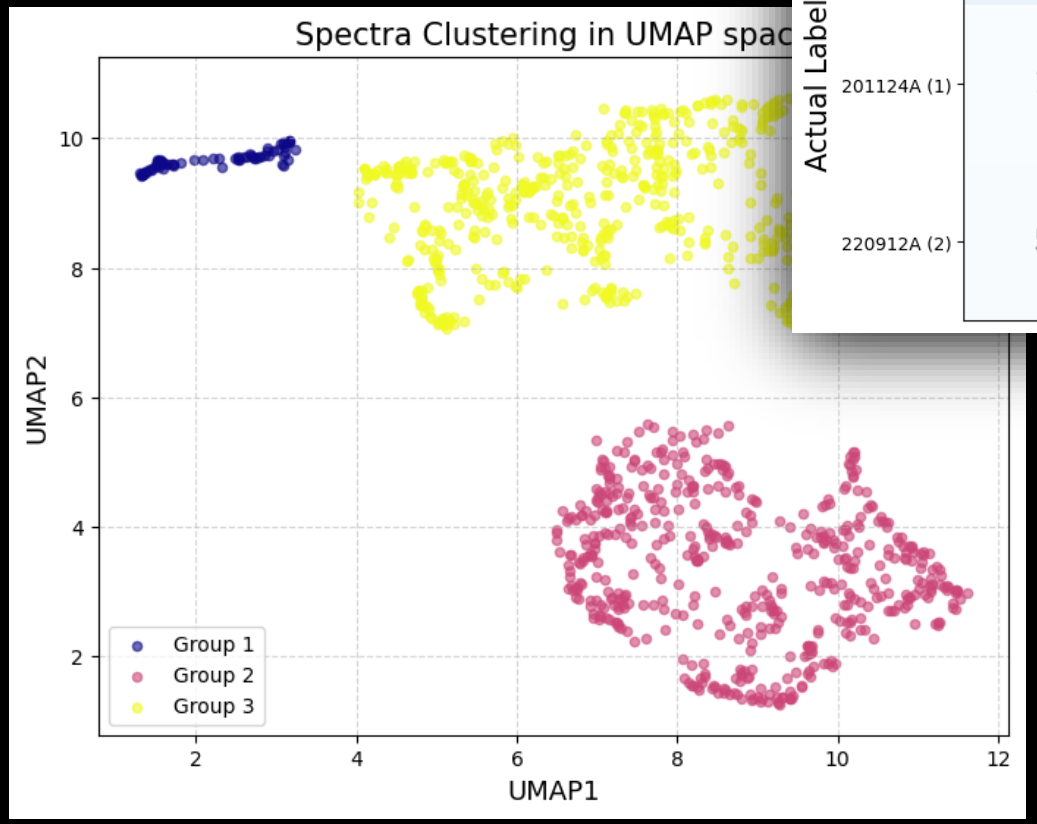


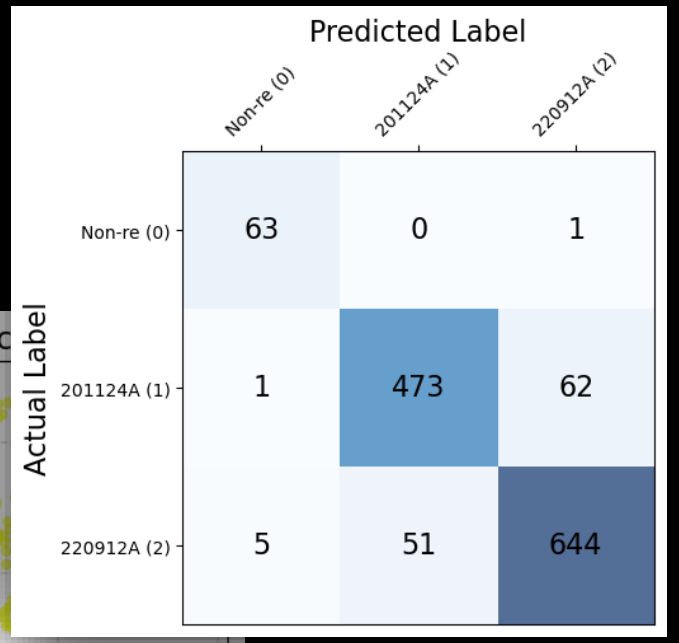


PC1 + PC2 explain 52% of total variance

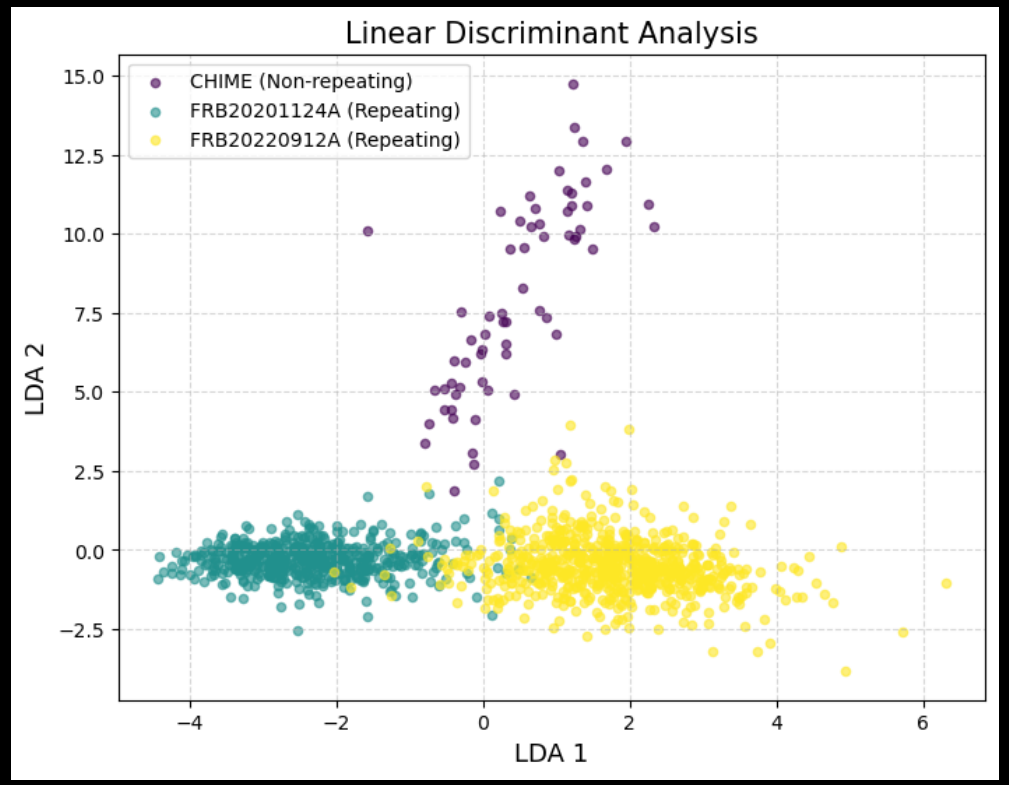
UMAP+Spectra Clustering

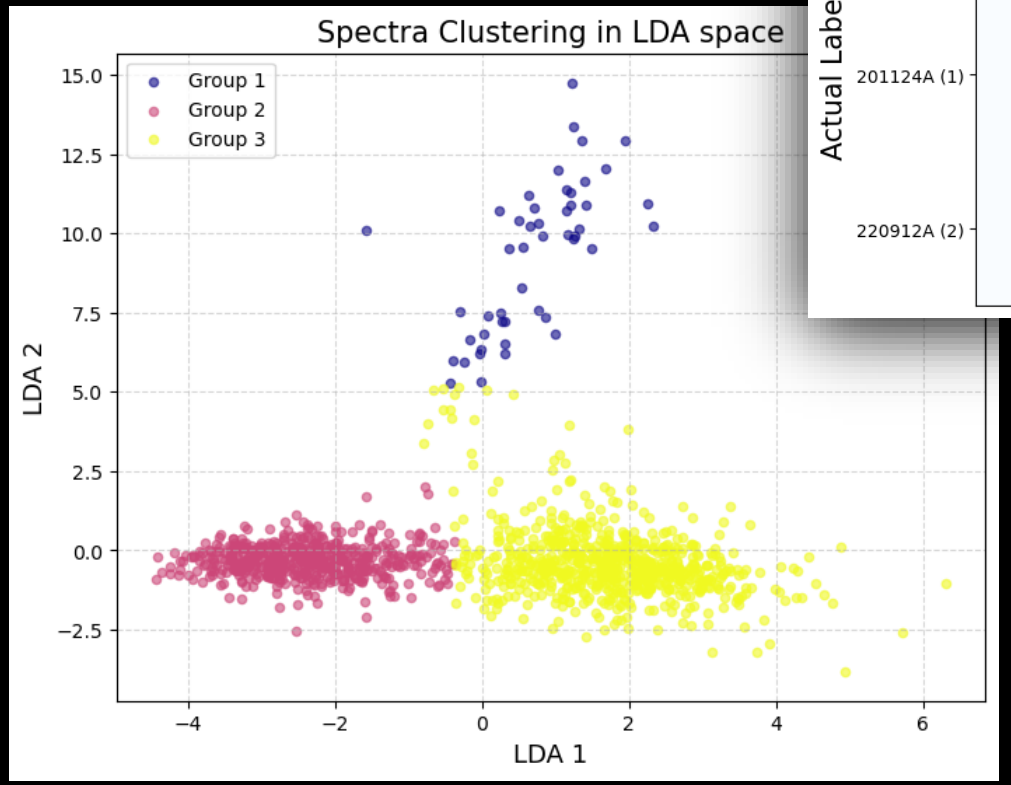


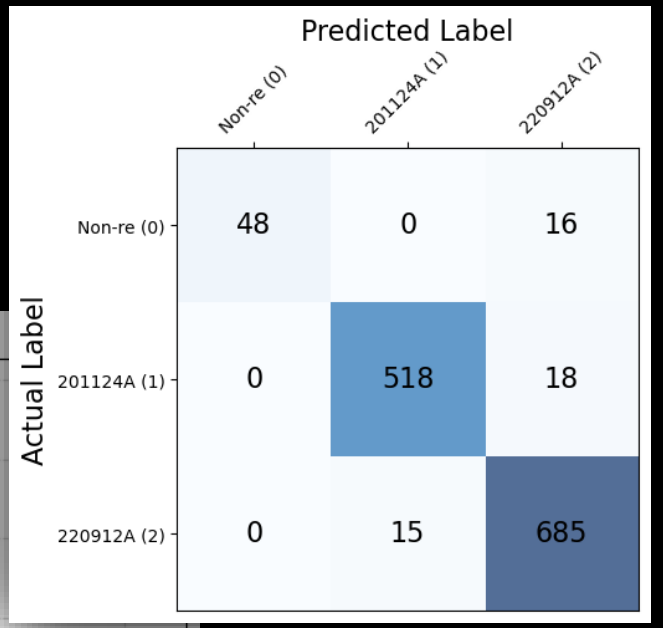




LDA+Spectra Clustering







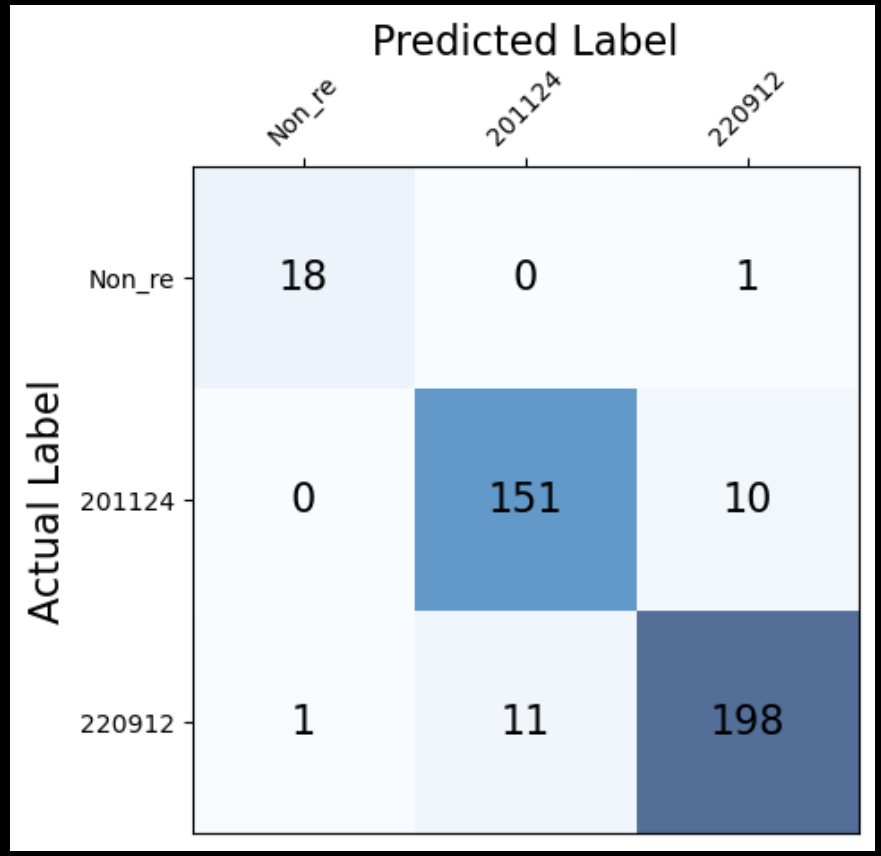
Classification Methods - XGBoost

	Original train-set	Processed train-set
Non-repeating FRB (0)	45	490
FRB 20201124A (1)	375	482
FRB 20220912A (2)	490	482

Smote + Tomek

	Precision	Recall	F1-score
Non-repeating FRB (0)	0.95	0.95	0.95
FRB 20201124A (1)	0.93	0.94	0.93
FRB 20220912A (2)	0.95	0.94	0.95

Classification report



Discussion

Polarization:

- Spectra describe burst morphology
- Polarization probes the magnetized environment
- Their combination provides complementary information

Result:

- Repeaters share similar latent features
- Non-repeaters occupy a distinct feature space
- Multimodal features improve discriminability

Limitations & Future Work:

- Only 2 repeating FRB sources
- Physical interpretation of latent features remains unclear

THANKS!